13.5.5 Physical Properties:

- Benzene is a colourless liquid, insoluble in water and solution alcohol, ether and chloroform.
- It burns with luminous sooty flame in contrast to alkanes and alkenes which usually burn with bluish flame.
- Their vapours are highly toxic which on inhalation produce loss of consciousness.

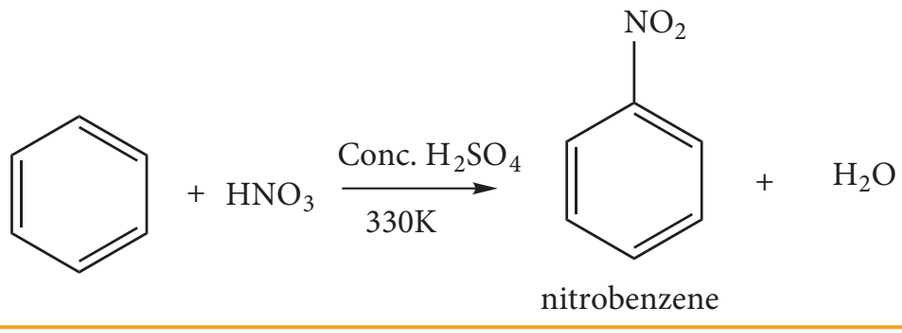
13.5.6 Chemical Properties:

1. Benzene contains delocalized π-electrons which make the ring to act as an electron rich centre. So electrophilic substitution reaction occurs in benzene.
2. Benzene ring is stabilized by delocalized π electrons. Though it is highly stable, it undergoes addition and oxidation reaction under specific conditions.

1. Electrophilic Substitution Reaction

(a) Nitration:

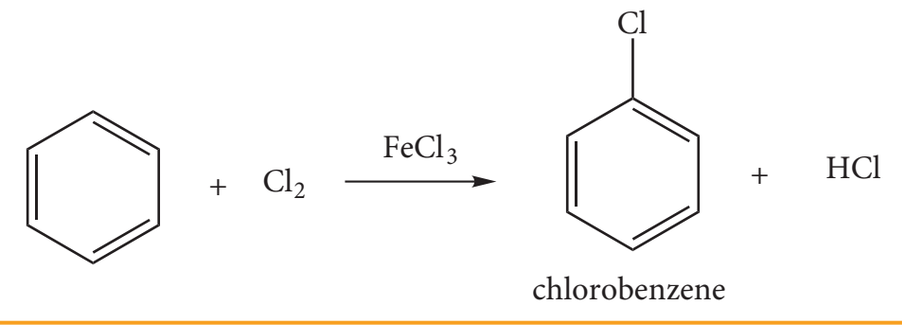
When benzene is heated at 330K with a nitrating mixture (Con. HNO_3 + Con. H_2SO_4), nitro benzene is formed by replacing one hydrogen atom by nitronium ion NO_2^+(electrophile)



Concentrated H_2SO_4 is added to produce nitronium ion NO_2^+

(b) Halogenation:

Benzene reacts with halogens (X_2=Cl_2, Br_2,) in the presence of Lewis acid such as $FeCl_3$, $FeBr_3$ or $AlCl_3$ and give corresponding halo benzene. In the absence of catalyst, Fluorine reacts vigoursly with benzene even in the absence of catalyst. However iodine is very inactive even in the presence of catalyst



(c) Sulphonation:

Benzene reacts with fuming sulphuric acid (Con H_2SO_4 + SO_3) and gives benzene sulphonic acid. The electrophile SO_3 is a molecule. Although it does not have positive charge, it is a strong electrophile.

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